

Vishesh Dubey

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Education

Indian Institute of Information Technology, Kurnool

Kurnool, Andhra Pradesh

Bachelor of Technology in Computer Science and Engineering; **CGPA: 9.1/10.0**

August 2024 – June 2028

Projects

OpsManager — Full Stack Manpower Management System with AI Assistant

June 2026

| *React, FastAPI, PostgreSQL, LangGraph*

- Built a full-stack manpower management platform spanning **7 interconnected PostgreSQL tables** for employees, customers, contracts, shifts, attendance, and incidents with relational foreign-key constraints.
- Implemented a **3-path LangGraph AI agent** that dynamically routes queries between pgvector-based semantic retrieval, natural-language-to-SQL generation, and direct conversation, with persistent multi-turn memory.
- Evaluated the AI assistant on **20 domain-specific queries**, with **10/10 RAG queries** producing relevant results and approximately **9/10 structured SQL queries** answered correctly.
- Engineered a defense-in-depth SQL safety layer using prompt constraints, **16 write/DDL operation checks**, statement-stacking prevention, and a dedicated read-only PostgreSQL role.
- Developed a FastAPI backend with **30+ REST API endpoints**, SQLAlchemy ORM, and Pydantic validation; implemented JWT authentication, secure password hashing, and protected React/Tailwind routes.
- Deployed the backend on Render and frontend on Vercel; [Live Demo](#) | [Frontend GitHub](#) | [Backend GitHub](#).

Custom Autograd Engine and Neural Network Library | *Python*

January 2026

- Built a scalar-valued automatic differentiation engine and neural network library from scratch, implementing topological sorting for reverse-mode backpropagation.

Technical Skills

Languages: Python, C++, C, JavaScript, SQL (PostgreSQL), HTML/CSS

Frontend: React, Tailwind CSS, React Router

Backend: FastAPI, SQLAlchemy, Pydantic, REST APIs, JWT Authentication

Database: PostgreSQL, pgvector, Supabase, Relational Database Design, ORM

AI/ML: LangGraph, LangChain, RAG, Cohere Embeddings, LLMs, Vector Databases

Tools: Git, GitHub, Linux (WSL2), VS Code, Render, Vercel

Certifications

Deep Learning for NLP | *NPTEL, IIT Kharagpur* — *Elite + Topper*

Jan – Apr 2026

- Scored **90%** and ranked in the **top 1%** of 731 certified candidates for a 12-week proctored course.

Supervised Machine Learning | *DeepLearning.AI, Coursera*

Dec 2025

- Completed a **33-hour** course with a **100% grade** implementing Regression and Gradient Descent.

Relevant Coursework

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| • Data Structures | • Database Management | • Theory of Computation | • Probability & Statistics |
| • Algorithms Analysis | • Artificial Intelligence | • Linear Algebra | • Linux |

Honors & Achievements

Academic Standing: Ranked within top 5% of batch at IIIT Kurnool (CGPA: 9.1/10.0)

NPTEL Topper: Ranked top 1% nationally in Deep Learning for NLP out of 731 certified candidates